

Unit 3

① Objective of activity planning

Activity planning means identifying and organizing the task required to complete s/w project successfully.

Main objectives are:-

- identify all activities
- determine task sequence
- estimate time
- allocate resources
- assign responsibilities
- set milestone
- prepare project schedules
- identify dependences
- monitor project progress
- control delay & risk.

① Identify all activities

break project into small manageable tasks

② determine task sequence

decide which activities should be performed & which depend on others.

③ estimate time

determine how long each activities will take.

④ Allocate resources

assign developers, testers, tools, hardware, other resources to activities

⑤ Assign Responsibilities

decide who will be responsible for each activity.

(F) Set milestones
identify important checkpoints or completion points
in the project

(G) prepare project schedules
create a realistic timetable for completing activities

(H) Identify dependencies
understand relationship between different activities

2) Project Schedules

it is a plan that shows what activities have to be performed, when they will start and finish, how long they will take & who/resources are involved.

Main elements

activity → a task that needs to be completed

Duration → time required to complete activity

start date: → date when activity begins

finish date: → date when activity ends

dependency: → relationship b/w activities.

Resources: → person, team, tool or equipment required

milestone: → checkpoint in project

Common techniques

In SPM, project schedules can be represented using: →

- Gantt chart
- Network diagram
- PERT
- CPM

Gantt

A bar chart showing activities against time.

Network diagram

shows activities & their dependences.

PERT (program evaluation & review techniques)

used when activity duration are uncertain.

CPM (critical path method)

used to identify critical path - the sequence of activities that determine minimum project completion time.

③ Activity

it is a specific task or piece of work that must be performed to complete s/w project.

ex:- suppose we are developing college management system:-

<u>activity</u>	<u>what it means</u>
requirement gathering	SDLC phase
system design	"
coding	"

each of these is an activity in the project.

An activity generally has:-

- name
- duration
- resource
- start/finish date
- predecessor
- successor

④ Sequencing & Scheduling

these activity help the project manager decide what should be done first, what comes next, and when each activity should be completed.

Sequencing

process of arranging project activities in correct order based on their dependencies.

ex:- Suppose we are developing online shopping application:-
requirement analysis → design → coding → testing

↓
deployment.

So, requirement analysis come before design & so on, this is called sequencing

Activity Dependency

a dependency means that one activity is related to or dependent on another activity.

ex:- coding → testing (testing depends on completion of coding)

Common dependency types

- FS (Finish-to-start)
- SS (Start-to-start)
- FF (Finish-to-finish)
- SF (Start-to-finish)

FS :- (Finish-to-start)
a most common dependency.

ex:- activity A must finish before activity B can start.

Design → coding

design must finish before coding starts.

SS (Start-to-start)

activity B can start when A starts.

ex:- start DB development → start backend development

both activity can begin around same time.
FF (finish-to-finish)

FF (finish-to-finish)

Activity B can't finish until A finishes

ex

Development $\rightarrow$ Documentation

SF (start-to-finish)

Activity B can't finish until A starts, less common.

Parallel activities

not every activity needs to wait for another activity.

for example

ample
after requirement analysis, two activities might begin:-

Requirements — [DB design
UI design] — development.

DB design & UI design can performed in parallel.

Sequencing example

<u>Activity</u>	<u>Description</u>	<u>Predecessor</u>
A	Requirement analysis	—
B	system design	A
C	DB design	B
D	UI design	B
E	coding	C, D
F	testing	E
G	deployment	F

Sequence is :-

$$A \rightarrow B \rightarrow C/D \rightarrow E \rightarrow F \rightarrow G$$

- Scheduling
- It determines when each activity will start and finish.
 - Sequencing → what comes next
 - Scheduling → when it will happen

ex:-

Suppose

- analysis → 50
- design → 40
- coding → 100
- testing → 50

A schedule might be:-

<u>Activity</u>	<u>Duration</u>	<u>Start</u>	<u>Finish</u>
Analysis	→ 5 Day	→ Day 1	→ Day 5
Design	→ 4 Day	→ Day 6	→ Day 9
coding	→ 10 Day	→ Day 10	→ Day 19
testing	→ 5 Day	→ Day 20	→ Day 24
	<u>24 Days</u>		

Steps

- Identify Activities
- estimate activity duration
- identify dependencies
- sequence activities
- assign resources
- identify critical path
- Monitor & update

Techniques

- Gantt chart
- Network Diagram
- PERT
- CPM

5) Network Planning Model (NPM)

- it is a graphical technique used in SPM to represent activities of project & dependencies between them.
- Network planning shows which activity comes first, which activity comes next, which activity runs together and how the entire project flow from start to finish

Basic Components of NPM

• Activity

an activity is a task that consumes time & resources.

ex:-

coding $\rightarrow$ 10 days

• Event/Milestone

An event represents the start or completion of an activity.

ex:-

"design completed"

• Dependency

it shows that one activity depends on another.

ex:-

A $\rightarrow$ B i.e. Design $\rightarrow$ coding, coding can't begin until design completed.

Types of NPM

• Activity on Arrow (AOA) $\rightarrow$ used in PERT

• Activity on Node (AON) $\rightarrow$ used in CPM

AOA

an arrow represents activity & node represents events.

ex:-

(1) $\xrightarrow{A}$ (2)

(2) $\xrightarrow{B}$ (3)

} A = Activity A
B = Activity B

May require dummy activity to represent certain dependency.

AON

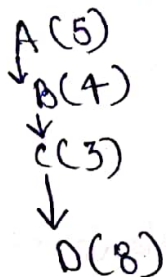
node represent an activity and arrow represent dependencies.

ex:- Requirement $\rightarrow$ Design $\rightarrow$ Coding
it is commonly used with modern projects.

Critical path

it is longest sequence of dependent activities in a project network and determines minimum time required to complete project.

ex:-



total $5+4+3+8 = 20$ days.

if this is longest path, project require at least 20 days

float/slack

amount of time an activity can be delayed without delaying project completion date.

ex:-

- Activity A has 0 days float $\rightarrow$ critical
- Activity B has 3 days float $\rightarrow$ can be delayed by 3 days

⑥ Formulating network model

it means converting activities of a s/w project into a logical network diagram that shows order dependencies, relationship among activities.

Why do we need to formulate network Model?

A network model helps project manager understand :-

- What activities must be performed?
- What activities should come first?
- Which activity depends on other activity? etc

Steps

- i> Identify all activities
- ii> Assign Activity ID
- iii> Estimate Activity Duration
- iv> Identify dependencies
- v> Determine parallel Activities
- vi> Choose Network representation
- vii> Draw Network
- viii> Add start & finish points
- ix> Check Network Logic
- x> Calculate Project Schedule

ES:- activity start without violating its predecessor dependency.

EF:- earliest time an activity finish based ES & duration.

LS:- activity can start without delaying overall project completion.

LF:- latest time an activity can finish without delaying the overall project completion.

Network logic

- Rule 1:- No activity is missing
Rule 2:- Dependencies are correct
Rule 3:- No unnecessary dependency
Rule 4:- No circular relationship.

Project Schedule

Important calculation include:-

- ES (earliest start)
- EF (earliest finish)
- LS (latest start)
- LF (latest finish)
- Float/Slack

3) Forward pass & Backward pass

Forward pass used to calculate:-

- Earliest start (ES)
- Earliest finish (EF)

Formula:-

$$EF = ES + \text{Duration}$$

ex:- if A activity start on Day 5 & takes 4 days:-

$$EF = 5 + 4 = 9$$

∴ earliest finish is Day 9.

Backward pass calculates:-

- latest finish (LF)
- latest start (LS)

Formula

$$LS = LF - \text{Duration}$$

these value help determine how much an activity can be delayed without affecting project completion date.

④ Float/slack

Amount of time an activity can be delayed without delaying project.

$$\text{Float} = LS - ES$$

or

$$\text{Float} = LF - EF$$

if $\text{Float} = 0$ [critical activity]

⑧ Rules for Network formulation

- i> every activity must have unique identification.
- ii> activity must be arranged according to their dependences.
- iii> Network should flow logically from [start $\rightarrow$ finish]
- iv> Avoid circular dependences.
- v> parallel activity should separately.
- vi> An activity should not start before its required predecessor are completed.
- vii> All activity eventually lead to project finish.
- viii> Duration should be associated with activities.
- ix> After network construction, calculate critical path & float.
- x> Network should clearly show relationship between activities.

⑨ CPM

critical path method is project scheduling technique used to determine the minimum time required to complete project & identify activities that are critical to project schedule.

Why do we use CPM?

CPM helps project manager to :-

- calculate minimum project completion time
- identify critical activities.
- monitor project progress.
- prioritize resources

Important terms

- Activity:- task that requires time & resources
- Duration:- time required to complete activity.

- predecessor
an activity that must be completed before another activity can start

- path
sequence of connected activities from start to finish

- critical path
longest duration path through network

- float/slack
amount of time an activity can be delayed

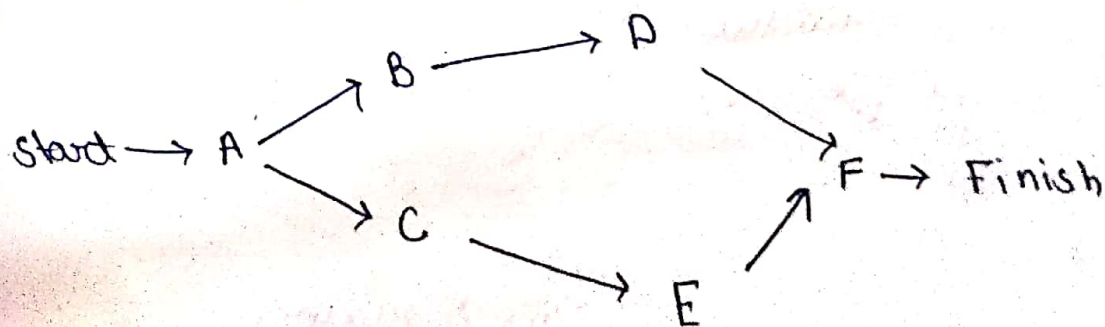
without delaying overall project.

⇒ Do CPM analysis and identify critical path of project.

<u>Activity</u>	<u>Predecessor</u>	<u>Duration (In month)</u>
A	—	5
B	A	4
C	A	5
D	B	6
E	C	3
F	D, E	4

Step 1 :- start CPM analysis from activity with no predecessors.

Step 2 :- Draw a network diagram using AON or activity on node representation.



- Step 3 :- For every activity find
- ES (earliest start)
 - EF (earliest finish)
 - LS (latest start)
 - LF (latest finish)
 - Duration (D)
 - Slack time (SL)

General Format

ES	D	EF
Activity		
LS	SL	LF

For A activity

0	5	5
A		
0	0	5

$$\begin{aligned} \text{Slack} &= \text{LF} - \text{EF} \\ &= 5 - 5 \\ &= 0 \end{aligned}$$

$$\min(5, 7) = 5$$

For B activity

5	4	9
B		
5	0	9

$$\begin{aligned} \text{Slack} &= \text{LF} - \text{EF} \\ &= 9 - 9 \\ &= 0 \end{aligned}$$

For C activity

5	5	10
C		
7	2	12

For D Activity

9	6	15
D		
9	0	15

For E Activity

10	3	13
E		
12	2	15

For F Activity

15	4	19
F		
15	0	19

$$\begin{aligned} \max(D, E) &= \max(13, 15) \\ &= 15 \end{aligned}$$

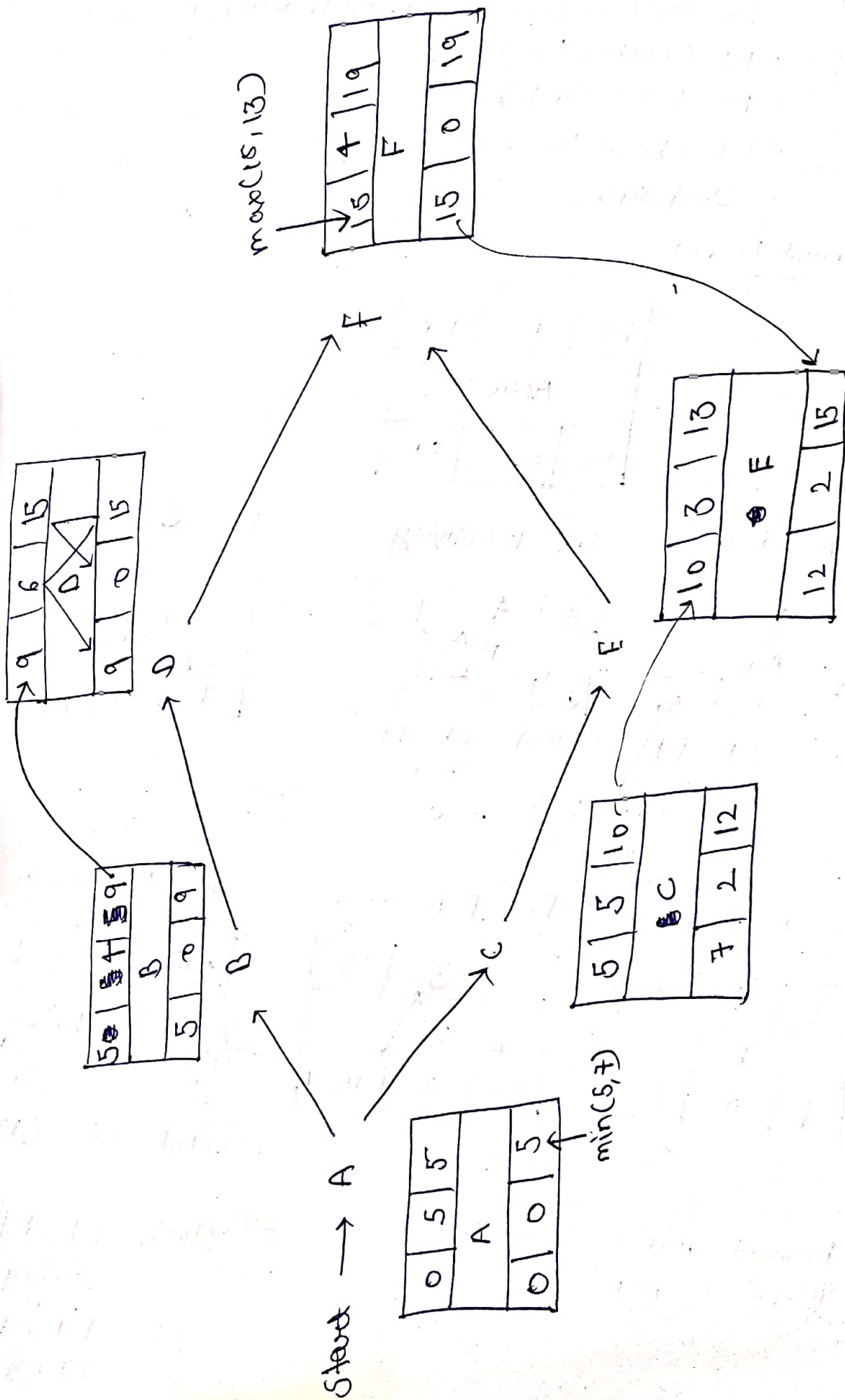
Forward = max
Backward = min

$$\text{Slack} = \text{LF} - \text{EF}$$

$$= 19 - 19 = 0$$

$$\text{LS} = \text{LF} - \text{Duration}$$

$$= 19 - 4 = 15$$



Possible path

(I) $A \rightarrow B \rightarrow D \rightarrow F$

(II) $A \rightarrow C \rightarrow E \rightarrow F$

Step 4 :- select that path in which slack time is 0.

$A \rightarrow B \rightarrow D \rightarrow F$ Duration =

(16) PERT

Program evaluation & review technique is a project scheduling technique used when duration of activity is uncertain.

It is especially useful in software projects because it is often difficult to predict exactly how long activities such as development, test, debugging or integration will take.

It is network-based project planning & scheduling technique that uses three time estimates - optimistic, most likely and pessimistic - to calculate expected duration of an activity.

Why Needed

Suppose developers is asked :-

"How many days will this feature take?". It may not be possible to give one exact answer. Developer might say best case is 4 day, Normal case is 6 day, Worst case is 10 days, PERT uses all 3 estimates instead of relying on only one

- 3 time estimates in PERT, is used 3 estimates
- i> optimistic time (O)
 - ii> Most likely time (M)
 - iii> Pessimistic Time (P)

O → Minimum time required when everything goes well.

M → Normal or most realistic time required.

P → Pessimistic time, maximum reasonable time when problem occur

PERT expected time formula

$$T_E = \frac{O + 4M + P}{6}$$

PERT standard deviation

$$\sigma = \frac{P - O}{6}$$

PERT variance

$$\sigma^2 = \left(\frac{P - O}{6} \right)^2$$

PERT steps

- identify activities
- determine dependencies
- estimate O, M, P
- calculate expected time
- construct network
- find critical path
- estimate project completion time

Conclusion

• E & L

E is earliest event occurrence time, it is earliest time at which that event can occur, calculate using forward pass.

L is latest event occurrence time, latest time at which that event can occur without delaying project, calculated using backward pass

• Activity expected time ($\bar{x}_e$)

$$\bar{x}_e = \frac{O + 4M + P}{6}$$

$$\sigma^2 = \left(\frac{P - O}{6} \right)^2$$

$$\sigma = \frac{P - O}{6}$$

• Standard deviation formula

it tells the range of uncertainty, if P and O are close $\rightarrow$ less uncertainty, if P and O are very far apart more uncertainty.

$$\sigma = \frac{P - O}{6}$$

• Variance

$$\sigma^2 = \left(\frac{P - O}{6} \right)^2$$

Project expected duration

$$T_E = \sum T_E \text{ of critical path activity}$$

Project standard deviation

$$\sigma_{\text{project}} = \sqrt{\sum \sigma_i^2}$$

Z-value formula

what is $P(E)$ that project will finish within n days.

$$Z = \frac{T_D - T_E}{\sigma_{\text{project}}}$$

T_D = desired / deadline time

T_E = expected project duration

σ_{project} = project standard deviation

Conclusion

Step 1 expected time

$$T_E = \frac{O + 4M + P}{6}$$

Step 2 standard deviation

$$\sigma = \frac{P - O}{6}$$

Step 3 variance

$$\sigma^2 = \left(\frac{P - O}{6} \right)^2$$

Step 4 critical path

Step 5 Project expected duration

$$T_E (\text{project}) = \sum T_E \text{ on critical path}$$

Step 6 project standard deviation

$$\sigma_{\text{project}} = \sqrt{\sum \sigma_i^2}$$

Step 7 Probability

$$Z = \frac{T_D - T_E}{\sigma \text{ project}}$$

Step 8 — Z table

Q The time estimates (in week) for activities of a PERT network are :-

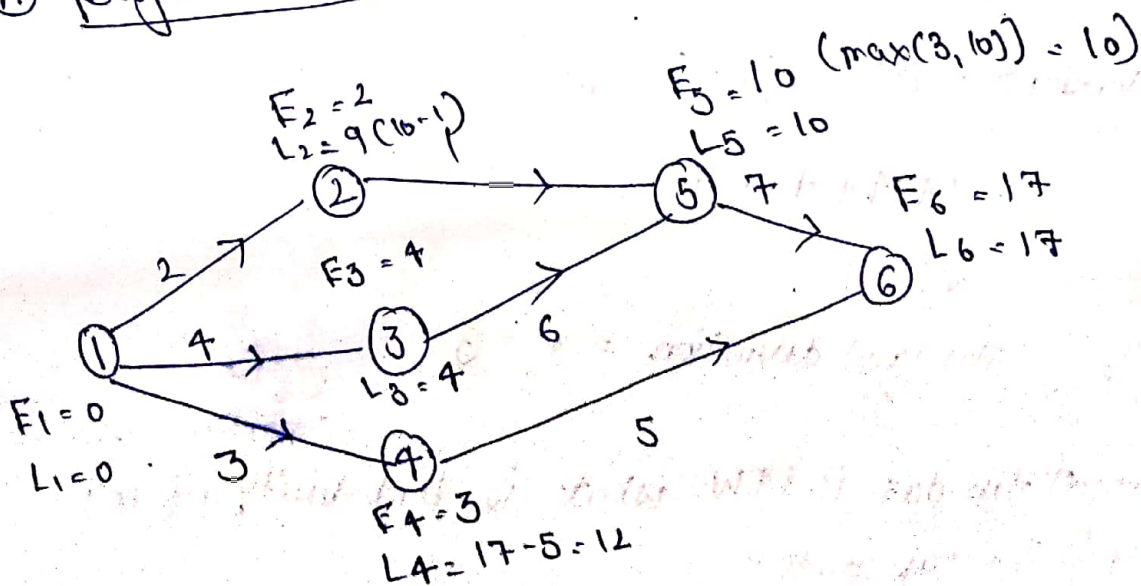
Activity	t_o (short, no delay, optimistic)	t_m (most likely)	t_p (pessimistic (long, most delay))
1-2 →	1	1	7
1-3 →	1	4	7
1-4 →	2	2	8
2-5 →	1	1	1
3-5 →	2	5	14
4-6 →	2	5	8
5-6 →	3	6	18

★ find expected project length

★ project network

★ standard deviation, variance of project length.

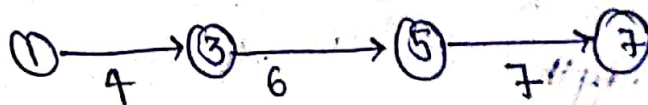
(A) project network



Activity	t_o	t_m	t_p	Expected time $t_e = t_o + 4t_m + t_p$ $\div 6$	Variance $\sigma^2 = \left(\frac{t_p - t_o}{6}\right)^2$
1-2	1	1	7	$t_e = \frac{1+4+7}{6} = 2$	$(7-1/6)^2 = 1$
1-3	1	4	7	$t_e = 4$	1
1-4 2-5	2	2	8	$t_e = 3$	1
2-5 3-5	1	1	1	$t_e = 1$	0
3-5 4-6	2	5	14	$t_e = 6$	4
4-6 5-6	2	5	8	$t_e = 5$	1
5-6	3	6	15	$t_e = 7$	7-4

Critical event is that event in which difference of EBL is 0 i.e. slack is 0

Critical path



Project length = $4 + 6 + 7 = 17$ weeks

Standard deviation (σ)

$$\text{Variance} = \sigma^2 = \sqrt{\sigma^2} = \sigma$$

$$= 1 + 4 + 4 = 9$$

$$\sigma^2 = 9$$

$$\therefore \text{Standard deviation} = \sigma = 3$$

Q If project due date is 19W, what is Probability of not meeting the due date?

Project due date = 19W

Expected time = 17W

Q. Aman is the dentist who schedule all his patient for 30 min appointment. Some of patient take more or less than 30 mins depending on type of work to be done. following Summarise shows various categories of work there probability and time actually needed to complete work.

category	time Required	No. of patients
Filling	45 min	40
Crown	60 min	15
cleaning	15 min	15
extracting	45 min	10
checkup	15 min	20

Simulate the dentist's clinic for 4 hours and find out dent average waiting time for patient as well as idleless of doctor.

Assume that all patient show up exactly at exactly there scheduling, &

Arrival time starting at 8:00 AM.

use the following random numbers for handling above problem.

(40, 82, 11, 34, 25, 66, 17, 79)

Range (11-82)

his

n 30

done.

ies

finally

s

step 1:- Establish probability distribution

step 2:- Establish Cumulating probability distribution

step 3:- Setting random number interval

step 4:- Generating random number.

steps:- To find answer of question ask using above four steps.

Solution Step 1:- find the probability distribution of all category.

$$\text{Probability} = \frac{\text{No. of outcome}}{\text{Total no. of outcome}}$$

category	Time Required	No. of Patients	Probability
filling	45 min	40	$\frac{40}{100} = 0.4$
crown	60 "	15	$\frac{15}{100} = 0.15$
cleaning	15 "	15	$\frac{15}{100} = 0.15$
extracting	15 "	10	$\frac{10}{100} = 0.10$
checkup	15 "	20	$\frac{20}{100} = 0.20$
		100	

Step 2:- find cumulative probability distribution

Cumulative probability:- It is the probability that of random variable will take on value less than or equal to specific cutoff.

$$F(x) = P(X \leq x)$$

* Probability ²⁰²⁰ & cumulative probability —

Probability means whether you are measuring the likely would of a single or exact event.

> cumulative probability is the running total of likely would up to a single point

example standard rolling dice.

A → what is the probability of rolling exactly of a 4:

$$P(X=4) = \frac{1}{6} \approx 16.7\%$$

B → what is the probability of rolling a number less than ^(at most) ~~at most~~ or equal to 4.

$$\begin{aligned} P(X \leq 4) &= P(X=1) + P(X=2) + P(X=3) + P(X=4) \\ &= \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{4}{6} \approx 66.7\% \end{aligned}$$

Type	Probability	Cumulative Probability
Filing	0.40	0.40
Crown	0.15	$0.40 + 0.15 = 0.55$
cleaning	0.15	$0.55 + 0.15 = 0.70$
extract	0.10	$0.70 + 0.10 = 0.80$
checkup	0.20	$0.80 + 0.20 = 1.00$

Step-3 "Set random number interval."

(First K' 0 to c. Probability - 1)

Type	Probability	Cumulative Probability	Random No.
Filing	0.40	0.40	00 - 39 (40-1)
Crown	0.15	0.55	40 - 54 (55-1)
Cleaning	0.15	0.70	55 - 69 (70-1)
Extract	0.10	0.80	70 - 79 (80-1)
Checkup	0.20	1.00	80 - 99

Step 4:- We set and generate random number

Patient	Scheduled Arrival	Random No.	Category	Service Time Needed
1	8:00	40	Crown	60 min
2	8:30	82	checkup	15 min
3	9:00	11	Filing	45 min
4	9:30	34	Filing	45 "
5	10:00	25	Filing	45 min
6	10:30	66	cleaning	15 "
7	11:00	17	filing	45 "
8	11:30	79	extract	45 "

Step 5:- we answer all questions (avg waiting time and idleness by Doctor) as per given table.

Patient	Arrival	Service start	Service duration	Service end	Waiting in min	Idle
1	8:00 AM	8:00 AM	60 min	9:00 AM	0 min	0
2	8:30 AM	8:00 AM	15 min	8:45 AM	30 min	0
3	9:00 AM	9:15 AM	45 min	10:00 AM	15 min	0
4	9:30 AM	10:00 AM	45 min	10:45 AM	30 min	0
5	10:00 AM	10:45 AM	45 min	11:30 AM	45 min	0
6	10:30 AM	11:30 AM	15 min	11:45 AM	60 min	0
7	11:00 AM	11:45 AM	45 min	12:30 PM	45 min	0
8	11:30 AM	12:30 PM	45 min	1:15 PM	60 min	0

Avg waiting time = $\frac{0 + 30 + 15 + 30 + 45 + 60 + 45 + 60}{8}$

$$= \frac{285}{8} = 35.62 \text{ min}$$

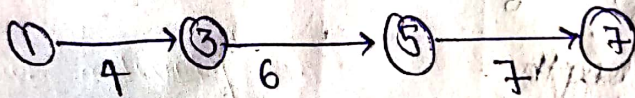
$$\approx 36 \text{ min}$$

Avg idle time = 0

Activity	t_o	t_m	t_p	Expected time $t_e = t_o + 4t_m + t_p$ 6	Variance $\sigma^2 = \left(\frac{t_p - t_o}{6}\right)^2$
1-2	1	1	7	$t_e = \frac{1+4+7}{6} = 2$	$(7-1/6)^2 = 1$
1-3	1	4	7	$t_e = 4$	1
1-4 2-5	2	2	8	$t_e = 3$	1
2-5 3-5	1	1	1	$t_e = 1$	0
3-5 4-6	2	5	14	$t_e = 6$	4
4-6 5-6	2	5	8	$t_e = 5$	1
5-6	3	6	15	$t_e = 7$	4

critical event is that event in which difference of EBL is 0 i.e. slack is 0

critical path



project length = $4 + 6 + 7 = 17$ weeks

standard deviation (σ)

$$\text{variance} = \sigma^2 = \sqrt{\sigma^2} = 6$$

$$= 1 + 4 + 4 = 9$$

$$\sigma^2 = 9$$

$$\therefore \text{standard deviation} = \sigma = 3$$

Q. If project due date is 19 W, what is probability of not meeting the due date?

Project due date = 19 W
expected time = 17 W

$$Z = P(X \leq 19) = \frac{X - \mu}{\sigma}$$

$$= \frac{19 - 17}{3} = \frac{2}{3} = 0.67$$

$$P = Z = 0.67$$

Now cumulative probability up to 0.67

$$\text{is } \psi(z) = 0.7486$$

or

$$74.86\%$$

$\therefore$ Z score tells where a specific data point stands compared to rest of a group.

$$P = 1 - \psi(z) = 1 - 0.7486 = 0.2514 = 25.14\%$$

Q The probability that project will be completed on schedule if scheduled completion time / is 20 weeks.

$$Z = \frac{X - \mu}{\sigma} = \frac{20 - 17}{3} = \frac{3}{3} = 1$$

$$P = 0.8413 / 84.13\%$$

Q What should be scheduled completion time for the probability of completion to be 90%?

$$P = 0.9$$

$$\psi(z) = 0.9$$

$$Z = 1.28$$

$$Z = \frac{X - \mu}{\sigma}$$

$$1.28 = \frac{X - 17}{3}$$

$$X = 20.84 \text{ weeks}$$

Note :- Z-score = How far am I from average

CPM Vs PERT

CPM

- suitable for construction project
- uses dummy activities for representing sequence of activities
- it has repetitive nature of jobs
- it has deterministic model
- it is activity oriented technique
- it is a technique of project management which is used to manage only certain activities of any project.

PERT

- suitable for project which require research & development.
- it doesn't use any dummy activity.
- It has non-repetitive nature of jobs
- it has a probability model
- it is event oriented technique
- it is a technique which is used to manage uncertain activities of any project.

① Risk identification, assessment, risk planning, risk management.

Risk

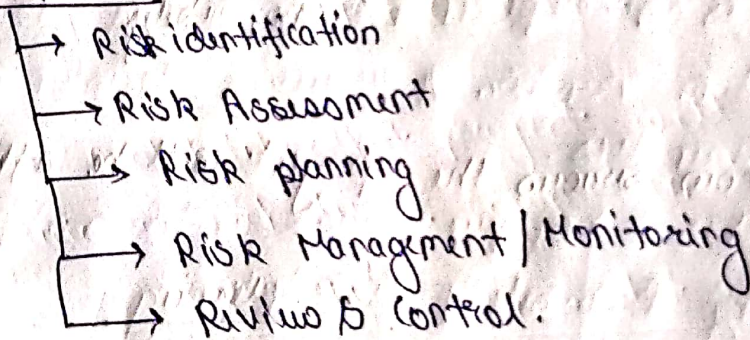
it is an uncertain event or condition that may occur in future. It has two important characteristics:-

- Uncertainty :- we don't know whether it will happen or not.
- Impact :- if it happens, it can affect project.

Risk Management

it is a systematic process of identifying, analyzing, planning response to, and monitor risks throughout a software project.

Main process



Risk identification

it is a process of finding & documenting potential risk that could affect the software project.

The project manager asks:-

"What can go wrong?"

Ex:- For an online shopping application:-

- developer may leave project
- requirement may change
- customer may delay approval.
- server may go down.
- 3rd party API may become unavailable.
- project may exceed budget.

Risk Assessment

After identifying risk, we need to determine:

1. How likely is the risk to happen?

2. How serious will its effect be?

Two important factors:-

Probability × Impact

Probability

→ it shows likelihood of a risk occurring.

Probability

- Low
- Medium
- High

Meaning

unlikely to happen
possible
very likely

Impact

Impact tells us how serious the project will be affected if risk occurs.

Impact

Low

Medium

High

Meaning

small effect

Noticeable effect

serious effect

Risk exposure / Risk Score

A simple quantitative approach is:-

Risk exposure = Probability × Impact

Risk planning

After assessment, project manager decides:-

"What should we do about this risk?"

this is called risk planning. The manager prepares

action to :-

• avoid risk

• reduce risk

• transfer risk

• accept risk.

Monte Carlo Simulation

- Simulation is process of designing a model of a real system & conducting experiment with model for purpose of understanding the behaviour for operation of system.
 - A duplication of original system.
 - to understand implementation of system.
- This method is used by various types of organization for 3 different uses.
like:-
 - a) as per different business situation
 - b) for inventory control
 - c) financial decision etc.
- This technique, is an experiment on chance (Probability)
- it uses random number & requires decision making under uncertainties.
- To understand this technique, break into 5 steps:-
 - I establishing Probability distribution.
 - II cumulative "
 - III setting random No. intervals -
 - IV Generating random no.
 - V to find answer of questions asked using above 4 steps.
- ⑥ A is a dentist who schedule all her patients for 30 minute appointment. Some of patient take more or less than 30 minute depending on type of dental work to be done. Following summary shows various categories of work, their probability and time actually needed to complete work.

of a real
for purpose
ation of system.

m.
of organization

situation

ce (Probability)
decision making

5 steps:-

using above 4

for 30 minutes

nan 30 minutes

showing summary
ility and time

Category	Time Required	No. of patient	Probability
Filling ✓	45 min ✓	40	$40/100 = 0.40$
Crown	60 min	15	$15/100 = 0.15$
Cleaning	15 min	15	$15/100 = 0.15$
Extracting	45 min	10	$10/100 = 0.10$
checkup	15 min	20	$20/100 = 0.20$
		<u>100</u>	<u>0.6</u>

$$PCE = \frac{\text{favourable outcome}}{\text{total outcome}}$$

Simulate the dentist clinic for 4 hrs and find out average wait
 time for patients as well as illness of the doctor. Assume
 that all patients show up at clinic at exactly their scheduled
 arrival time starting at 8:00 am.
 * Use the following random no. for handling above problem.

Solve table-I

Type	Probability	C. Probability	Random No.
Filling	0.40	0.40	00 - 39
Crown	0.15	$0.40 + 0.15 = 0.55$	40 - 54
Cleaning	0.15	0.70	55 - 69
Extracting	0.10	0.80	70 - 79
checkup	0.20	1.00	80 - 99

table-II

Patient	Scheduled Arrival	Random No. (column)	category	Service Time Required
1	8:00	40	Crown	60 min
2	8:30	82	checkup	15 min
3	9:00	11	Filling	45 min
4	9:30	84	Filling	45 min
5	10:00	25	Filling	45 min
6	10:30	66	cleaning	15 min
7	11:00	17	Filling	45 min
8	11:30	79	extracting	45 min

Q Use the following random no. for handling above problem.

40, 82, 11, 34, 25, 66, 17, 79 (serial wise)

Table-III (Final table)

patient	Arrival	Service start	Service Duration	Service ends	waiting in minutes	Idle time
1	8:00	8:00	60 min	9:00	0	0
2	8:30	9:00	15 min	9:15	30 min	0
3	9:00	9:15	45 min	10:00	15 min	0
4	9:30	10:00	45 min	10:45	30 min	0
5	10:00	10:45	45 min	11:30	45 min	0
6	10:30	11:30	15 min	11:45 min	60 min	0
7	11:00	11:45	45 min	12:30	45 min	0
8	11:30	12:30	45 min	1:15	60 min	0

$$\text{Average waiting time} = \frac{0 + 30 + 15 + 30 + \dots + 60}{8 \text{ (total patient)}}$$

$$= \underline{285} = 35.625 \text{ min.}$$

Idle time = 0 minutes

Theory

- this method is used to predict and understand the behaviour of systems involving uncertainty.
- By running multiple simulation with random input, this technique helps estimate possible outcome & their probability.

Application

- i. Reinforcement learning
- ii. uncertainty estimation
- iii. Risk assessment in financial models

Benefits

- Risk Analysis
- Better Decision making
- flexibility across fields
- Handling complex systems
- visualization of uncertainty.

Limitation

- computational intensity :- requires significant time & resources, specially with complex models.
- Quality of input data :- result depend on accuracy of input data, poor data leads to unreliable outcomes.
- convergence Rate :- higher accuracy often needs a large number of simulations, for rare events.
- over-simplification :- may miss complex system details or relationship, leading to potential oversight.